

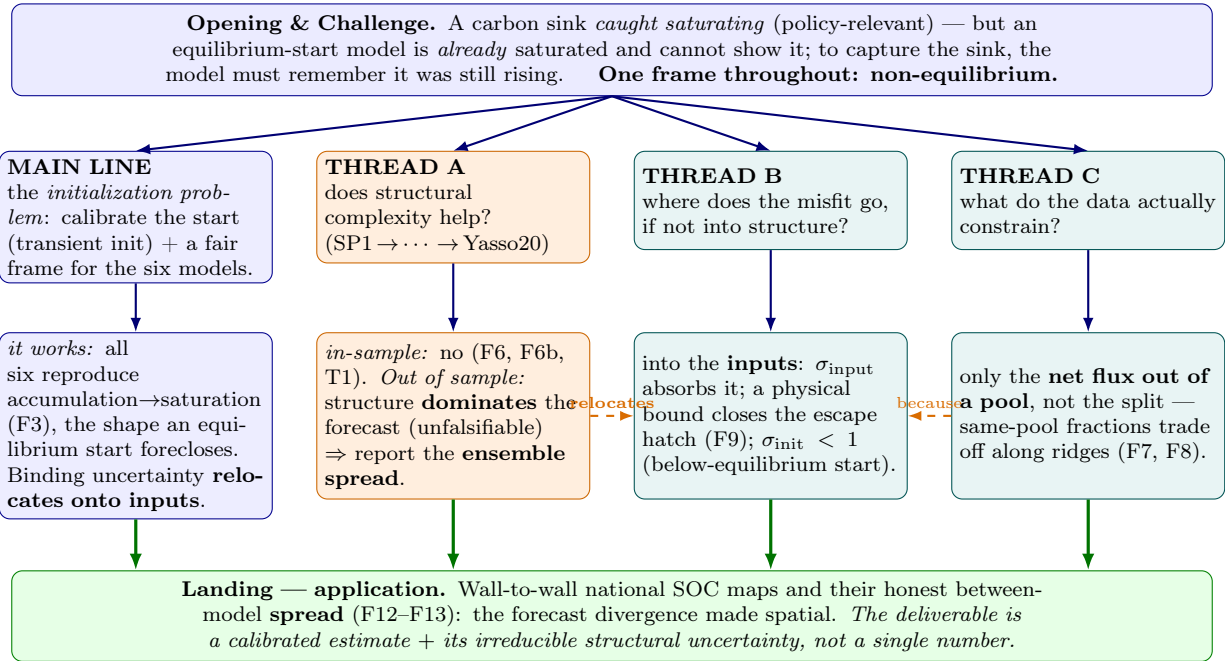
HIKET — the storyline

a working note for coauthors

Version of July 15, 2026 — a narrative testbed, not the manuscript. No methods or full referencing; the point is to read the argument whole and see how it flows.

What this is. Below is the HIKET argument told as one continuous story, with each figure and table placed where the story needs it. It follows an Opening–Challenge–Action–Resolution arc with three side threads; the schema on the next page is the map. The prose is deliberately compact — enough to feel the flow and judge whether it lands — and is the scaffold the manuscript will be written onto.

The story at a glance



The through-line is one physical fact — **these soils are not at equilibrium** — seen from two sides: it is why there is a sink to report (the stakes) and why a conventional model cannot show it (the obstacle). The whole paper is the work of turning the second into a solution for the first.

1 Opening: a sink caught saturating, and a model that cannot see it

Finnish forest soils have been accumulating carbon through the observation period and are now bending toward a plateau. This incipient saturation is not a modelling curiosity: Finland reports it in its national greenhouse-gas inventory, where most countries assume a flat soil pool, so getting it

right has direct policy weight. The signal has a physical cause. Across the twentieth century the country's forests recovered from a depleted, heavily-used early-century base, and the growing stock has risen by roughly seventy percent since the 1970s (Fig. 1); the litter those forests return to the soil rose with them and then levelled off, and it varies strongly from south to north (Fig. 2). A rising, non-stationary input history feeding soils that were well below their carrying capacity is exactly the setup for an accumulating, saturating sink.

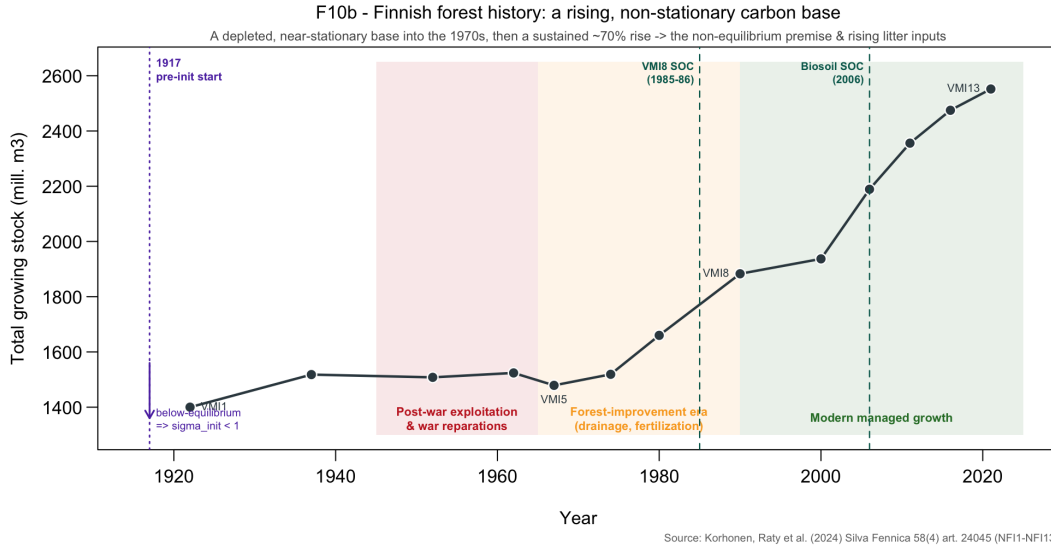


Figure 1: The non-equilibrium premise. National growing stock (NFI, 1921–2023): a depleted, near-stationary base into the 1970s, then a sustained recovery. The 1917 pre-initialisation anchor and the two SOC campaigns are marked. Source: Korhonen, Rätty et al. (2024).

The trap is that a soil model run the conventional way cannot represent any of this. Started at steady state — the near-universal default — the model already sits at its saturated end-state, with nowhere left to accumulate. Run forward at published parameters from an equilibrium start, it stays flat, and well below the observed stocks (Fig. 3). The very phenomenon we most want to capture is the one the standard convention forecloses.

2 Challenge: to show the saturation, the model must remember it was still rising

The stakes and the obstacle are the same fact from two sides. The soils hold an accumulating sink *because* they are out of equilibrium; the standard model fails *because* it assumes they are not. So capturing the observed dynamic is not a matter of better parameters — it requires abandoning the equilibrium-start convention and initialising the model as a system still recovering from its history. That is the initialization problem, and it is the axis the whole study turns on: how do you start a non-equilibrium simulation when you have no record of the forcing that produced the initial state?

3 Action: transient initialization under a fair frame

The move is to stop *assuming* the initial state and instead *calibrate* it. Each model is spun up from a below-equilibrium historical anchor to the start of the observation period, and the initial carbon

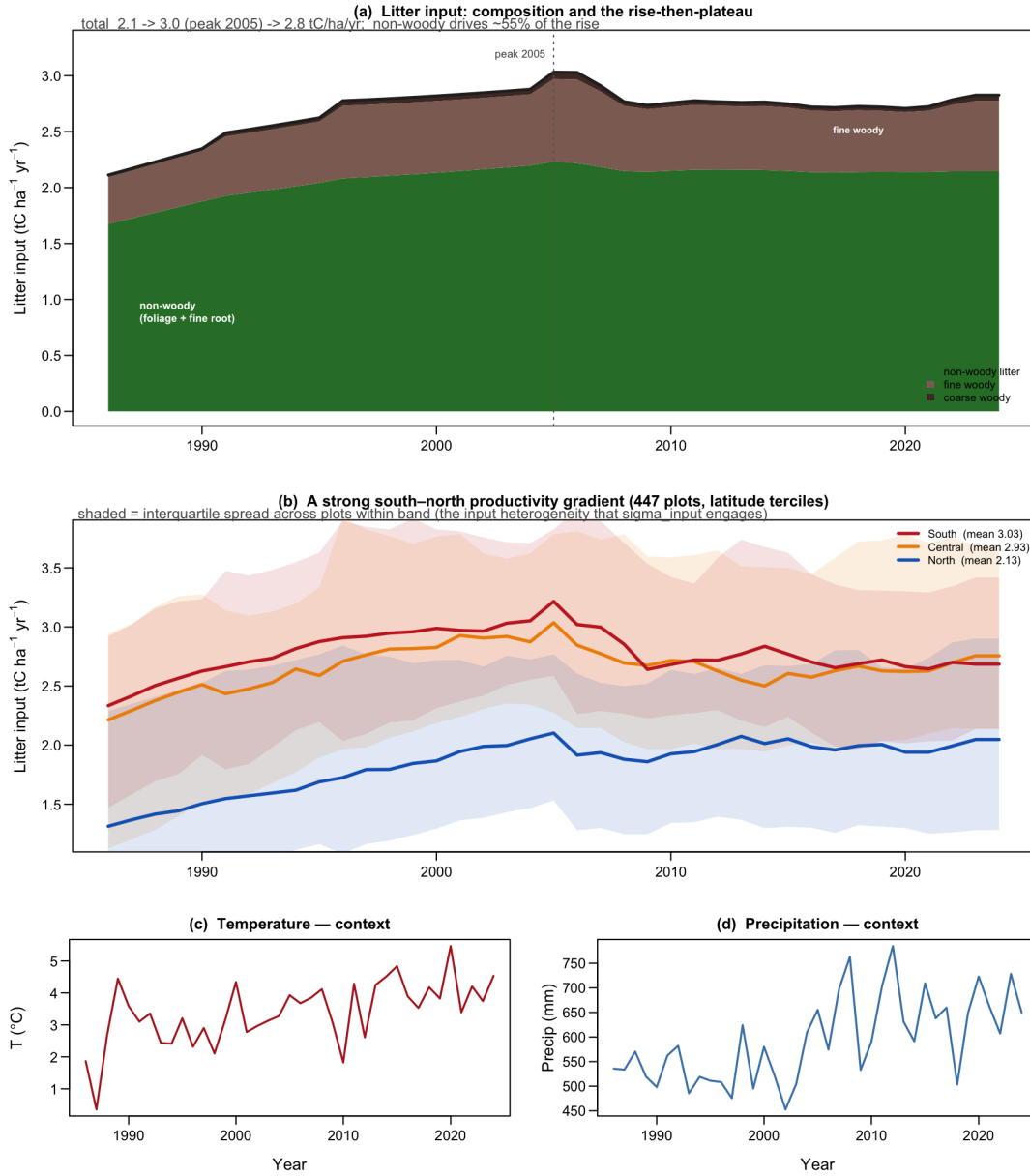


Figure 2: The inputs. Litter forcing over time (composition; a rise to a mid-2000s peak, then a plateau) and its strong south–north gradient, with climate shown only as context. The accumulation is input- and history-driven, and the inputs vary enormously in space.

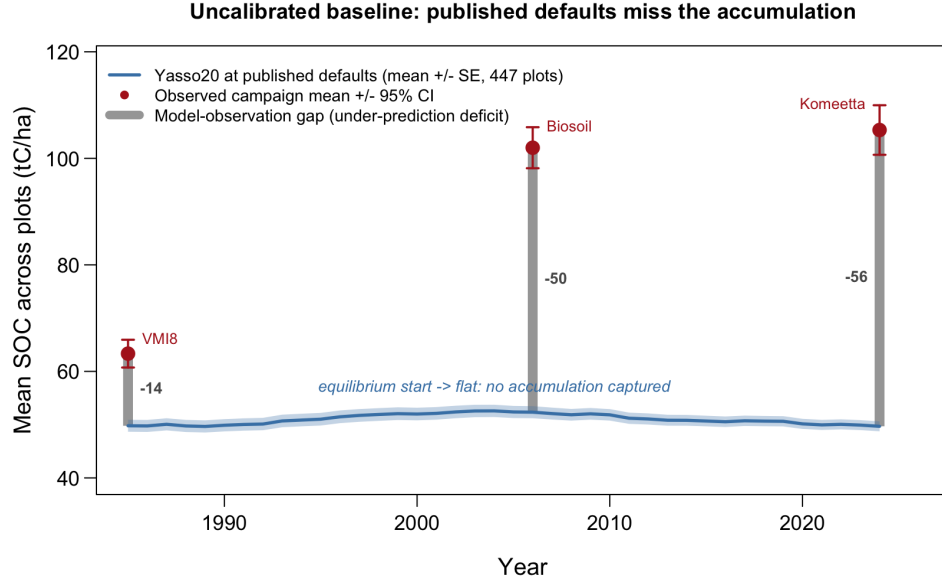


Figure 3: The trap. Yasso20 at published defaults with an equilibrium start, averaged over all plots: flat, because an equilibrium start cannot accumulate, and low relative to the observed campaign means.

becomes a calibrated quantity, σ_{init} , whose uncertainty propagates into both the starting state and the forward run rather than being fixed by fiat. Casting the unknown start as an inferred parameter, not a fixed assumption, is what makes this tractable at scale.

The comparison across models only means something if the models are not secretly competing on who guessed the inputs or the initial state best. We therefore put all six on an even footing: homogenised priors (identical in construction, not in raw number), equal degrees of freedom in the inter-pool flow fractions, and two auxiliary parameters — σ_{init} for the initial state and σ_{input} for the litter level — that absorb init- and input-uncertainty identically for every model. The six span a deliberate complexity ladder: a one-pool model (SP1), two- and three-pool cascades (TP2, TP3), and the three five-pool Yasso variants, with a second, climate-integration axis running Yasso07 $\rightarrow$ 15 $\rightarrow$ 20. On this even ground, structure — and only structure — is what varies.

4 Resolution: it works, and the binding uncertainty relocates

4.1 The payoff

Transiently initialised, all six models reproduce the observed shape — accumulation bending toward incipient saturation — and track the later campaign means (Fig. 4). This is the shape an equilibrium start cannot produce: solving the initialization problem is precisely what lets the models represent the policy-relevant sink. The payoff comes with an honest caveat visible in the reconstructed trajectory (Fig. 5): the models converge on a 1985 stock some 26 tC ha⁻¹ above the observed start, a consequence of the coarse, unrecorded pre-1985 phase — a live question, not a failure of the method.

4.2 Thread A — does adding structural complexity improve the answer?

In-sample, no. Predictive skill does not climb the pool ladder or the climate sub-axis; the most complex model fits worst, on both calibration and independent holdout (Fig. 6, Fig. 7). We do make the anti-complexity claim, but scope-bounded: for an applied, recalibratable, single-region task whose

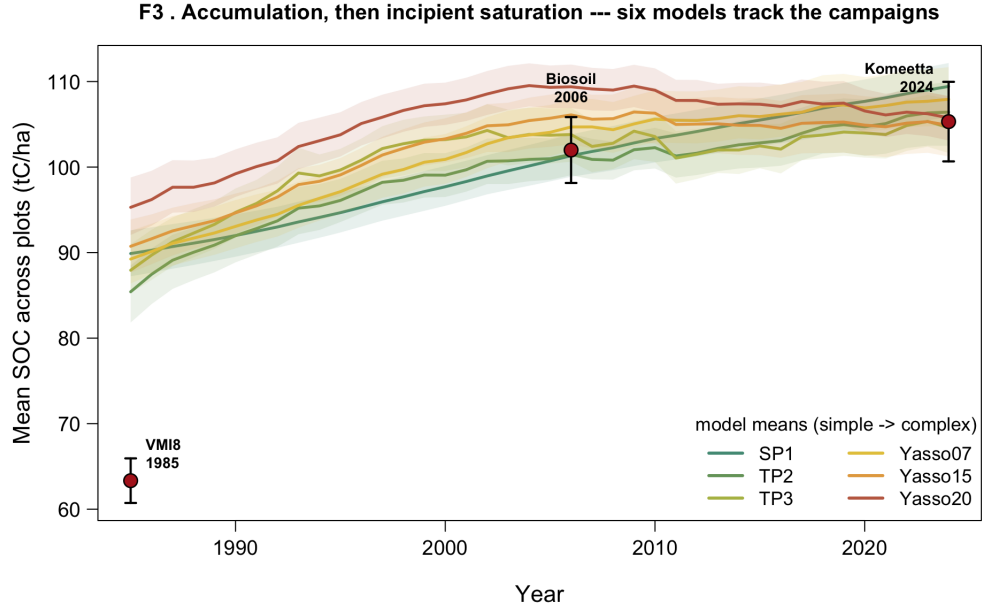


Figure 4: The central result. Cross-plot mean SOC, all six models, 1985–2024: rise then bend toward saturation, tracking the Biosoil/Komeetta campaign means. Colour runs from the simple models (green) to the Yasso family (warm).

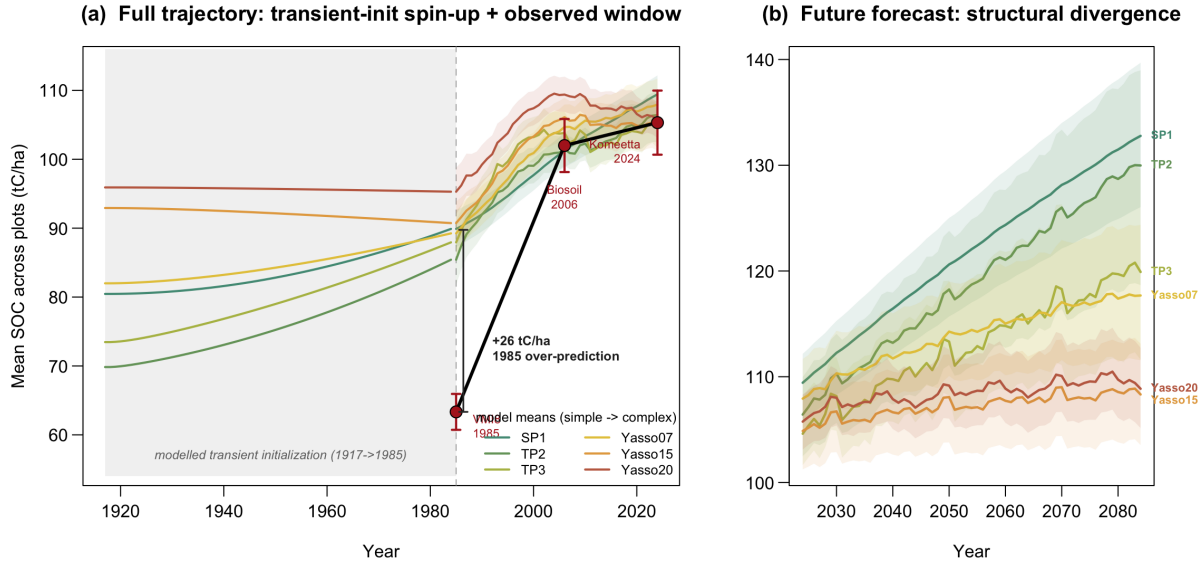


Figure 5: The initialization problem, made visible. (a) The full trajectory including the reconstructed transient-init spin-up 1917→1985 and the observed window; the models recover from a below-equilibrium start but disagree about how much, and all reach $\sim 26 \text{ tC ha}^{-1}$ above the observed 1985 stock. (b) The future forecast 2025→2084: agreement where calibrated, sharp divergence when extrapolated.

aim is accuracy plus calibrated uncertainty, added structure does not buy skill and can cost it. That is not a verdict on Yasso, which is optimised for a different objective — generality, standardisation, and litter-quality resolution that this single-region fit test simply does not measure — so “more complexity does not help *here*” and “Yasso is the right inventory tool” coexist. Table 1 is the whole comparison at a glance.

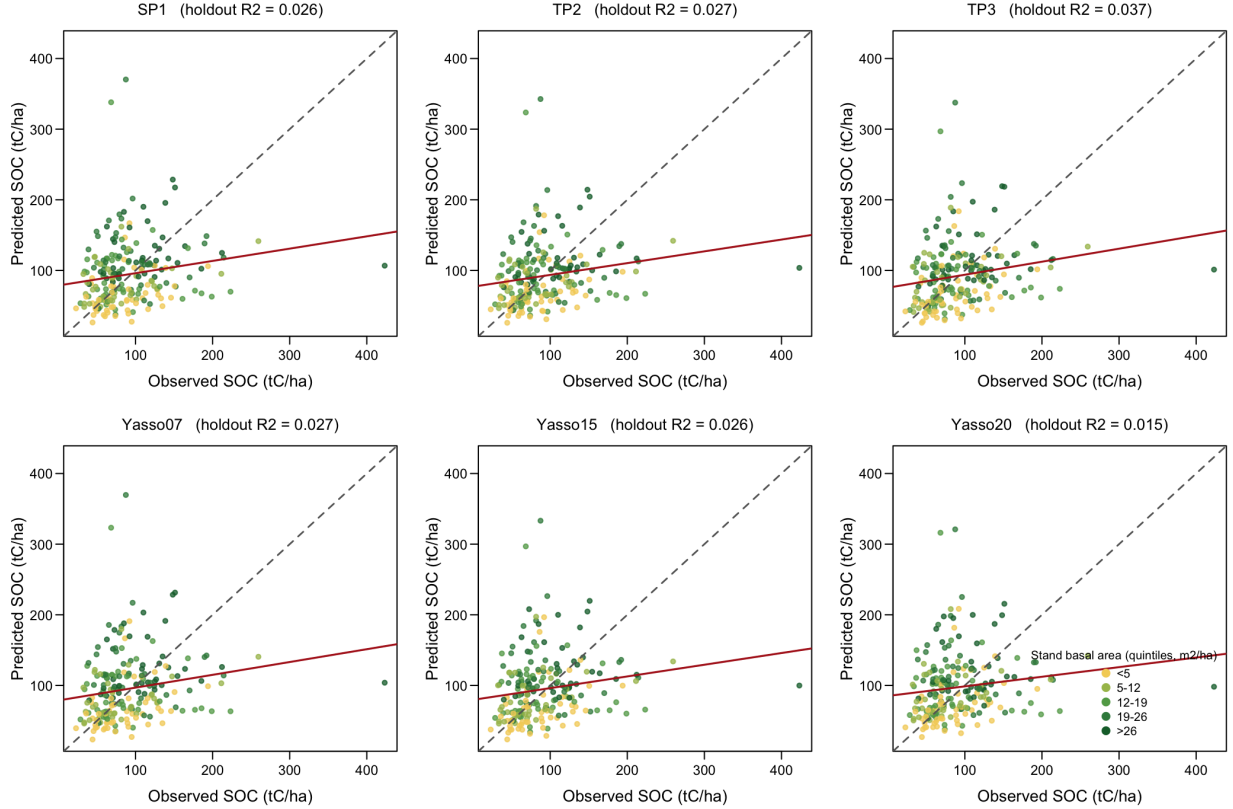


Figure 6: Skill on the honest set. Observed vs. predicted SOC on the held-out plots, by model, with points classed by stand basal-area quintile. Skill is uniformly low and the ranking nearly collapses out of sample; the red least-squares line is far flatter than the 1:1 dashed line (the models compress the range).

The picture inverts out of sample. Extrapolated to 2084, the models agree where the data constrain them and then fan out sharply — the one-pool SP1 climbs without bound toward $\sim 133 \text{ tC ha}^{-1}$, while the stiffer Yasso structures plateau near 108 (Fig. 5b). The very structural choices the data cannot pin down now govern the projection, and in-sample skill is blind to this because it is scored only on the present stocks. Process plausibility gives a soft steer — soils saturate, so SP1’s unbounded rise is the least credible trajectory — but it is soft, and the data cannot adjudicate. The decision-relevant quantity, multi-decade SOC change, is precisely the one in-sample skill cannot validate; the honest output is therefore the ensemble spread, not a single model.

4.3 Thread B — the binding uncertainty is the inputs

Where does the misfit go, if not into structure? Into the inputs. The multiplier σ_{input} is where each model absorbs what its kinetics cannot. Left as a weak, scale-free nuisance, the likelihood drags two models (TP2, TP3) to effective litter fluxes of $13\text{--}20\times$ boreal net primary production — physically impossible. Bounding the effective flux to a physical envelope closes that escape hatch, and the diagnostic is decisive: under the bound all six fall inside the envelope, and *only* the two runaway

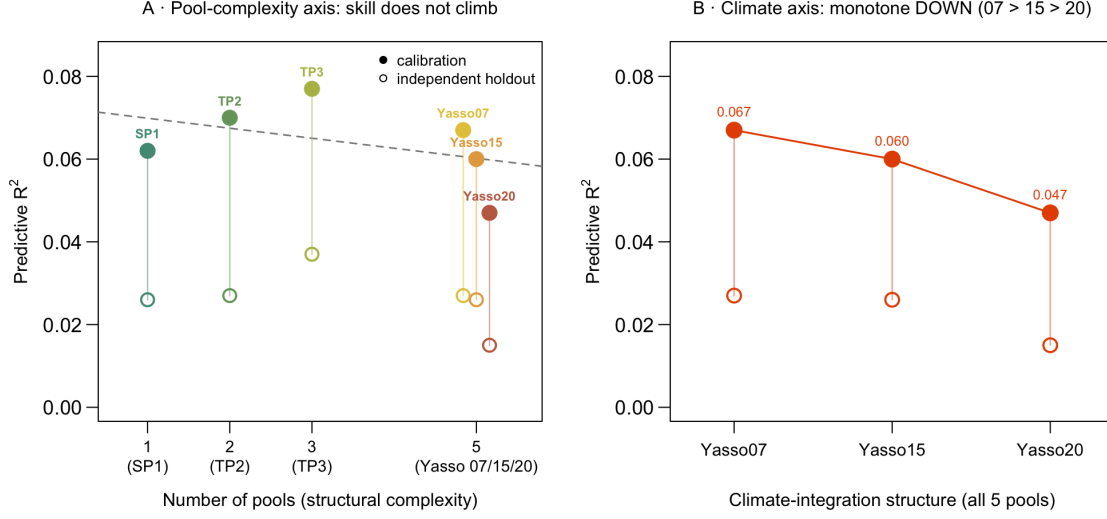


Figure 7: Complexity does not move the answer. Predictive R^2 against structural complexity: the pool-axis slope is negative and the climate sub-axis falls monotonically; the most complex model is the worst on both calibration and holdout.

Table 1: Results at a glance. Per-model structural complexity (pools, free parameters), skill (R^2 calibration and holdout, holdout RMSE), the input story (effective litter flux $\sigma_{\text{input}} \bar{J}$, all inside the physical $[0.5, 8.7]$ $\text{tC ha}^{-1} \text{yr}^{-1}$ envelope), and the initial state ($\sigma_{\text{init}} < 1$). Complexity climbs while skill does not.

Model	Pools	Free params	R^2 (cal)	R^2 (hold)	RMSE (hold)	Eff. flux $\sigma_{\text{inp}} \bar{J}$	σ_{init}
SP1	1	6	0.078	0.026	61	0.9	0.35
TP2	2	8	0.089	0.027	59	6.1	0.15
TP3	3	10	0.094	0.037	58	6.2	0.24
Yasso07	5	20	0.086	0.027	62	3.4	0.51
Yasso15	5	26	0.074	0.026	60	4.5	0.72
Yasso20	5	26	0.062	0.015	62	5.5	0.69

models lose predictive skill (Fig. 8). That the skill drop is *selective* — bounding a shared input nuisance singles out exactly the two models that had been manufacturing carbon — is the cleanest evidence that inputs, not kinetics, were the lever. The companion parameter σ_{init} is uniformly below one: a below-equilibrium start, the forest history of Fig. 1 made quantitative.

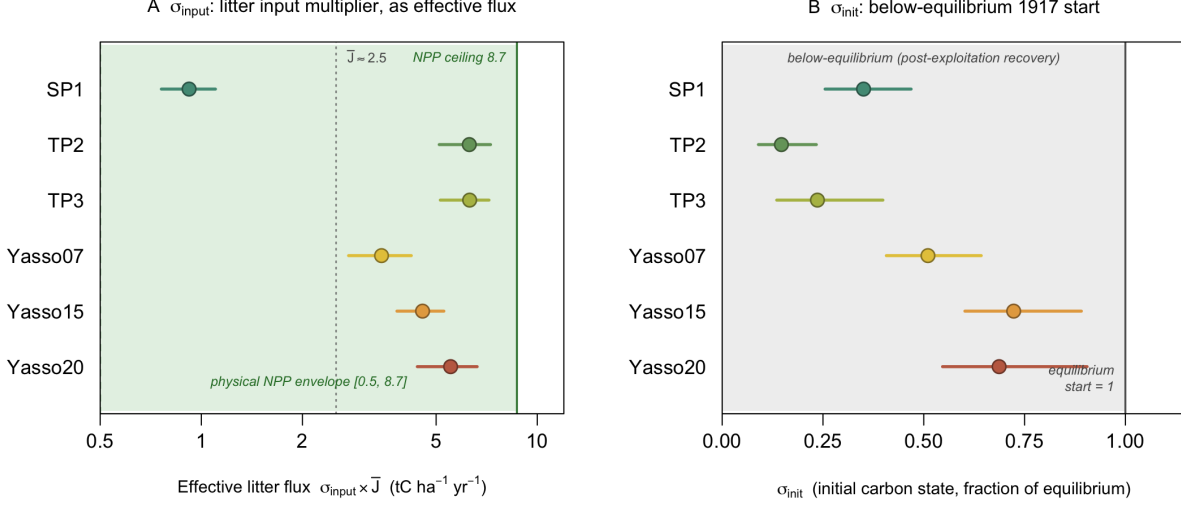


Figure 8: The auxiliary parameters, both physically interpretable. (a) σ_{input} as effective litter flux against the boreal NPP envelope (green); under the physical bound all six sit inside it. (b) σ_{init} as the initial carbon state; every interval lies below one, a post-exploitation, still-recovering start.

4.4 Thread C — what the data can and cannot constrain

The reason structure is a weak lever is that the data barely see it. The twelve inter-pool transfer fractions gain essentially no information from calibration — their posteriors stay on their priors (Fig. 9) — while σ_{input} gains the most. The mechanism is geometric (Fig. 10): fractions leaving the same pool anti-correlate almost perfectly, so the data constrain only the *net* flux out of each pool, never how it splits among destinations. Each source pool contributes one identified quantity; everything else is a ridge the posterior slides along (Table 2). This is the mechanism behind “complexity buys shape, not skill”: the parameters that complexity adds are the ones the data cannot resolve. We do not treat this as a defect to be fixed — it is intrinsic to the model class and the two stocks per plot — but manage it openly, pinning rates where authoritative values exist (Yasso) and calibrating them where they do not (the simple models).

Table 2: The strongest same-pool anti-correlations (Yasso20): each is a net-flux trade-off along which the posterior is unconstrained.

Source pool	Fraction pair	Posterior r	Interpretation
N	$p_{NA} \leftrightarrow p_{NE}$	-0.98	net-flux trade-off
E	$p_{EA} \leftrightarrow p_{EN}$	-0.86	net-flux trade-off
A	$p_{AW} \leftrightarrow p_{AN}$	-0.84	net-flux trade-off
A	$p_{AW} \leftrightarrow p_{AE}$	-0.81	net-flux trade-off
N	$p_{NA} \leftrightarrow p_{NW}$	-0.71	net-flux trade-off
W	$p_{WA} \leftrightarrow p_{WN}$	-0.37	net-flux trade-off

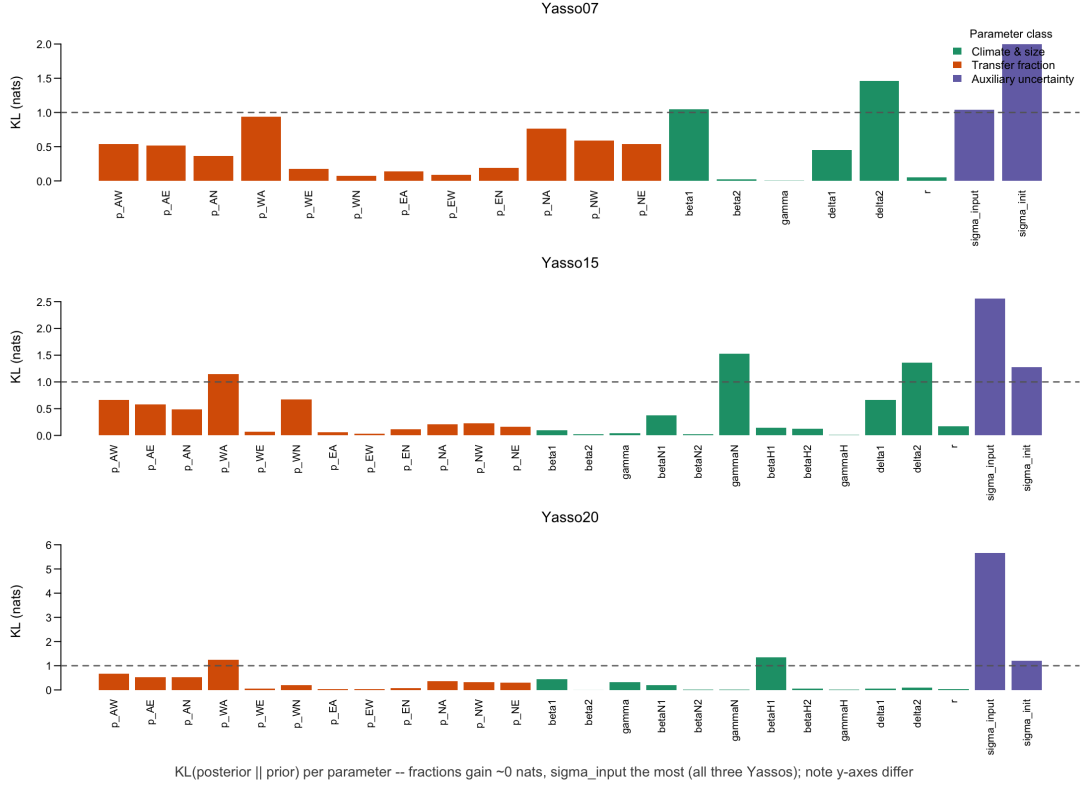


Figure 9: Information gain per parameter, the three Yasso models. The transfer fractions (orange) gain ≈ 0 nats; σ_{input} (purple) gains the most. The verdict — data constrain inputs, not kinetics — is the same across all three variants.

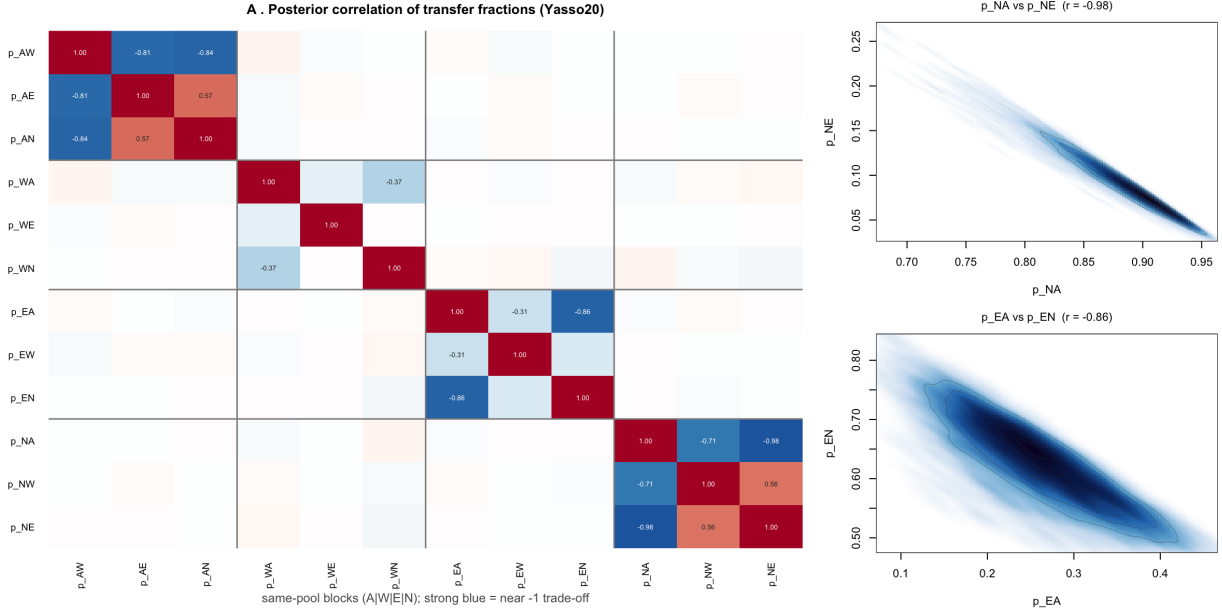


Figure 10: The trade-off structure (Yasso20). (a) Posterior correlations of the transfer fractions, blocked by source pool: same-pool fractions anti-correlate near -1 ; cross-pool correlations vanish. (b) Two same-pool ridges as continuous density fields. The data fix the net flux out of a pool, not the split.

4.5 The frontier: where the recoverable signal still hides

If not in the kinetics, where is the signal that remains? A random-forest analysis of the residuals answers the same way across every model: the leading, diffuse, shared driver is stand basal area — a productivity and input proxy, not a kinetic or pool variable (Fig. 11). What is left to explain lives in the forcing, not the kinetics, pointing in the same direction as everything above: the way forward is better inputs, not a bigger model.

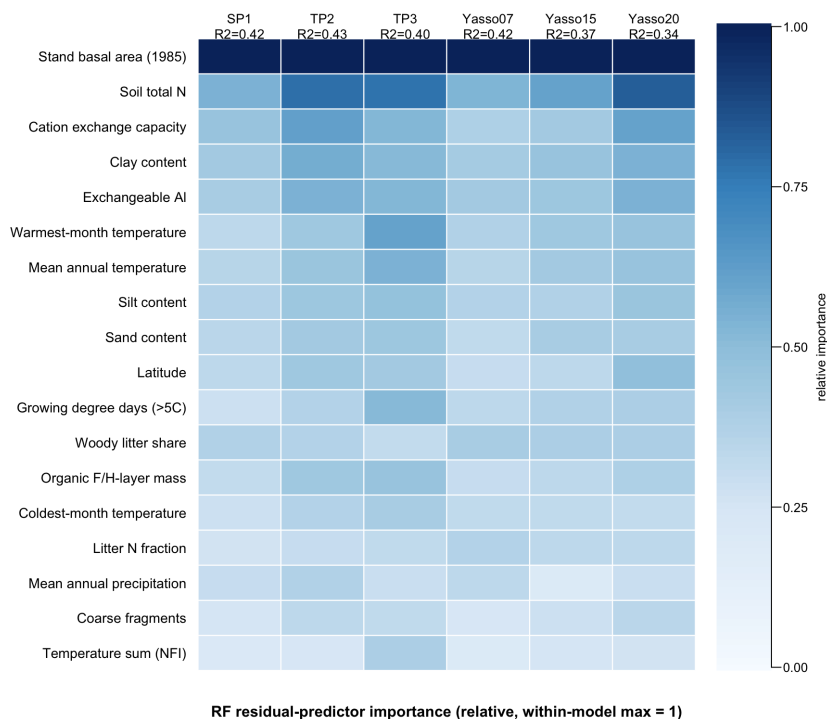


Figure 11: Residual structure. Random-forest importance of candidate drivers for the plot-level residuals (relative, within model). Stand basal area leads in every model; the residual points to site productivity and inputs, not to kinetics.



Landing: the policy deliverable

The application is a wall-to-wall national SOC product. The ensemble mean gives a country-wide stock estimate, and its between-model spread maps where the six structures agree and where they do not (Fig. 12) — the spatial face of the forecast divergence, and the reason the uncertainty panel is not an afterthought but the point. Alongside the current rate of change, the pool composition gives a second, distinct axis: how the carbon is held, from fast-cycling, less-processed material to stabilised humus (Fig. 13). What the study hands to the inventory is therefore not a single number but a calibrated estimate together with its irreducible structural uncertainty.

6 Outlook: closing the loop

The story ends where it began. Solving the initialization problem is what let us quantify the saturating sink; the residual disagreement about *how much more* it will saturate is the uncertainty we hand to

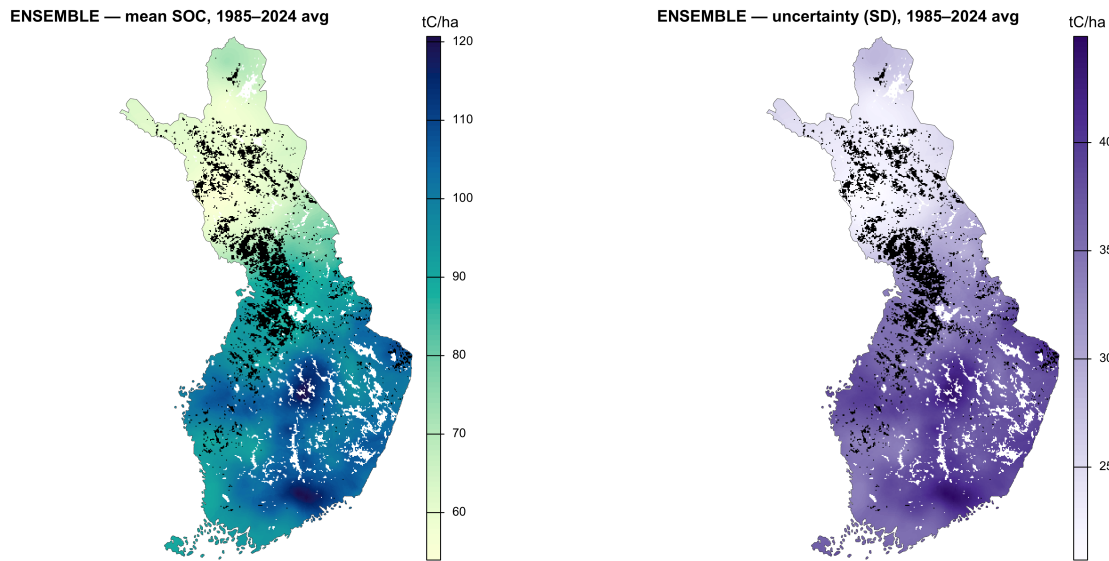


Figure 12: National SOC and its honest uncertainty. (a) Kriged ensemble-mean SOC over Finland; (b) the between-model spread per cell — modest but structured, the spatial expression of “structure is a weak lever.”

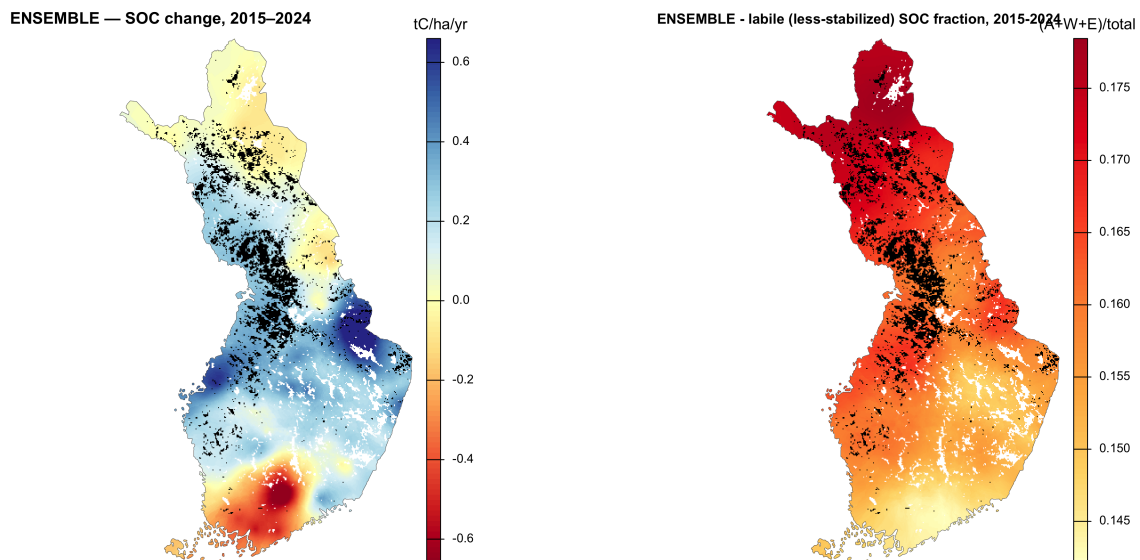


Figure 13: Two axes of soil-carbon dynamics. (a) The 10-year SOC change — gaining vs. losing soils; (b) the labile (less-stabilised) SOC fraction — how the carbon is held. The two are only partly correlated, so (b) adds a distinct signal rather than restating (a).

policy; and the way to narrow that uncertainty is not more model structure but better forcing — reconstructing the historical input trajectory (replacing our linear guess at the pre-observation phase) and, as an option, stratifying input level by site class. Non-equilibrium is the frame that ties the levels together: it is the cause (a managed forest history), the obstacle (an initial state we cannot assume), and the stakes (a sink still in motion) — one thing, seen from the model side and the policy side at once.

A Supporting figures

These are the supplement: robustness and companion figures that back the main claims without carrying the narrative. They are kept in full so no detail is lost relative to the working storyboard.

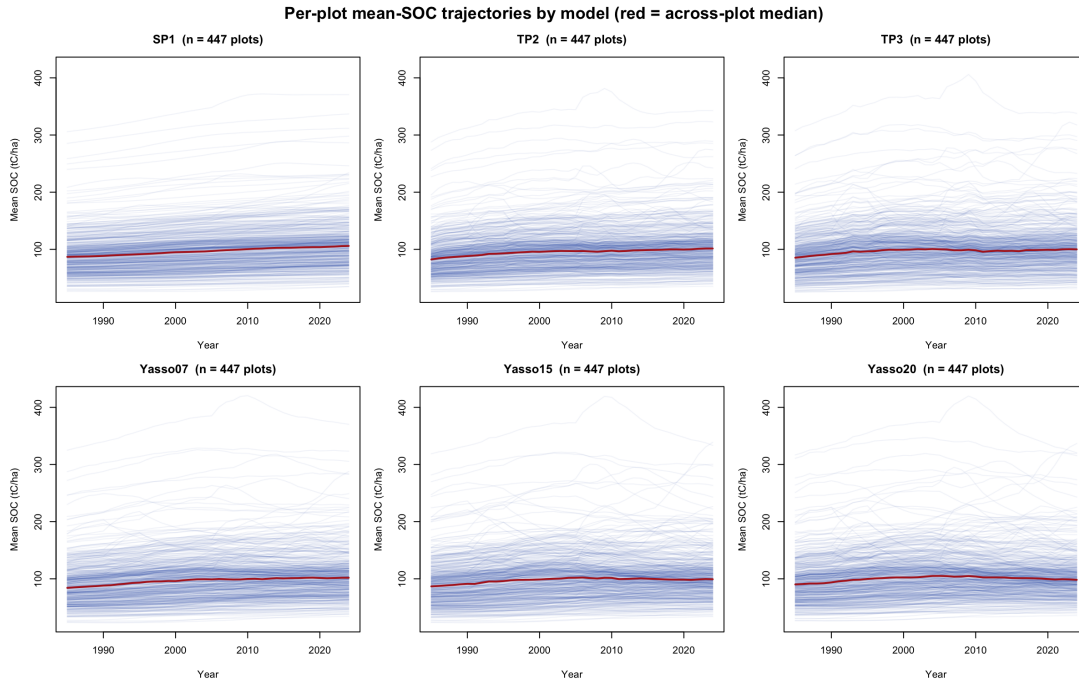


Figure 14: S1 — per-plot SOC trajectories. Every plot’s mean SOC by model; all now peak at a physical $\sim 100 \text{ tC ha}^{-1}$ (TP2/TP3 jagged, SP1 smooth). Confirms the flux bound produced no inflated stocks and that the TP3 oscillation is a real fast-pool signal, not a numerical artefact.

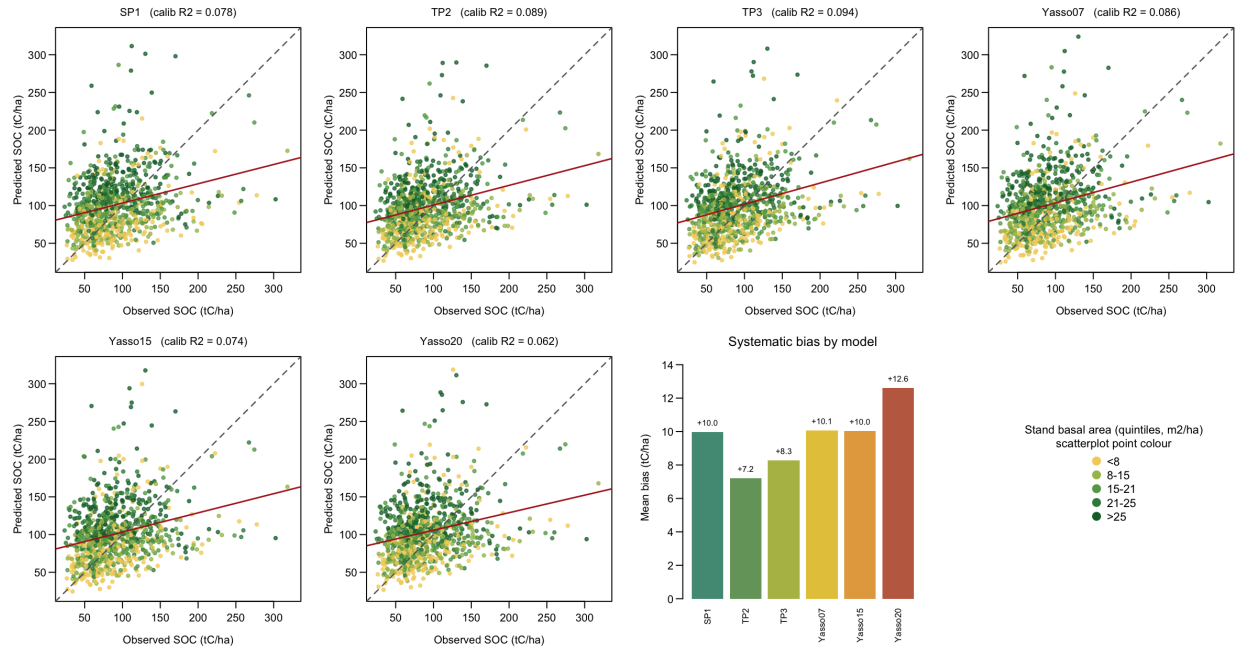


Figure 15: S2 — observed vs. predicted, calibration plots. The in-sample companion to Fig. 6 (six scatterplots coloured by basal-area quintile, as there) plus a systematic-bias barplot in the per-model palette. Same small spread, one notch higher than holdout; bias highest for Yasso20.

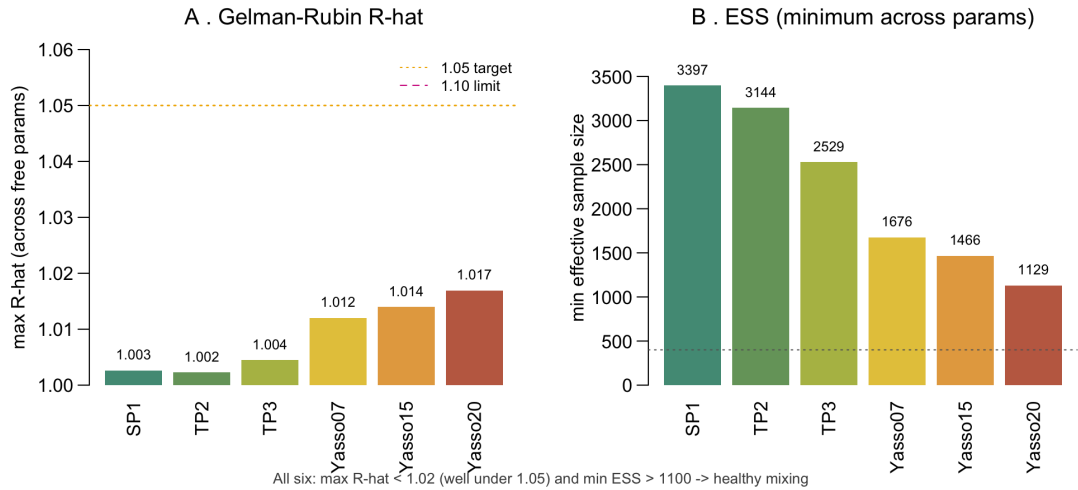


Figure 16: S3 — MCMC convergence. Per-model maximum R-hat (<1.02 everywhere, well under 1.05) and minimum effective sample size (>1100). Good mixing — so well-behaved fraction R-hat reflects mixing, not identifiability (the non-identifiability shows up in information gain and prior-pinning, not in R-hat).

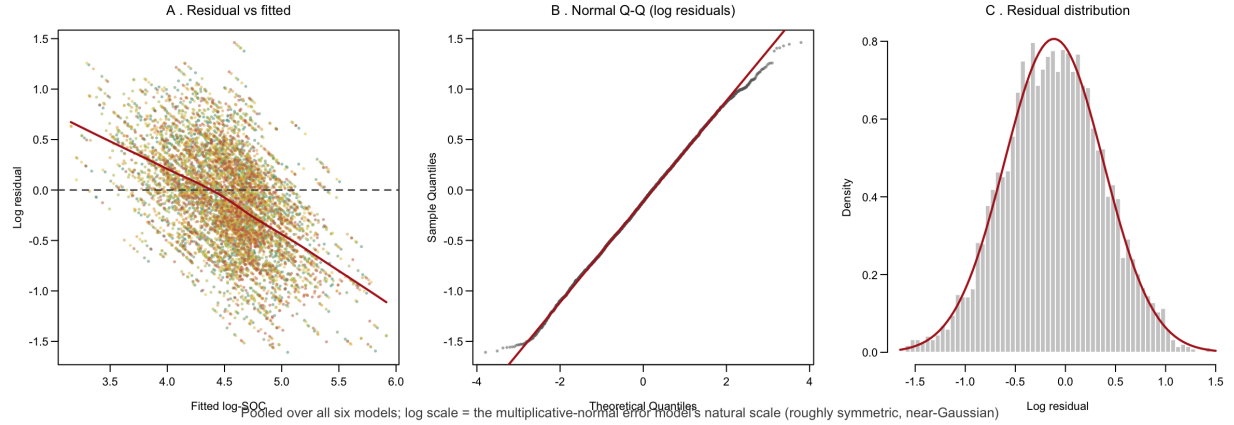
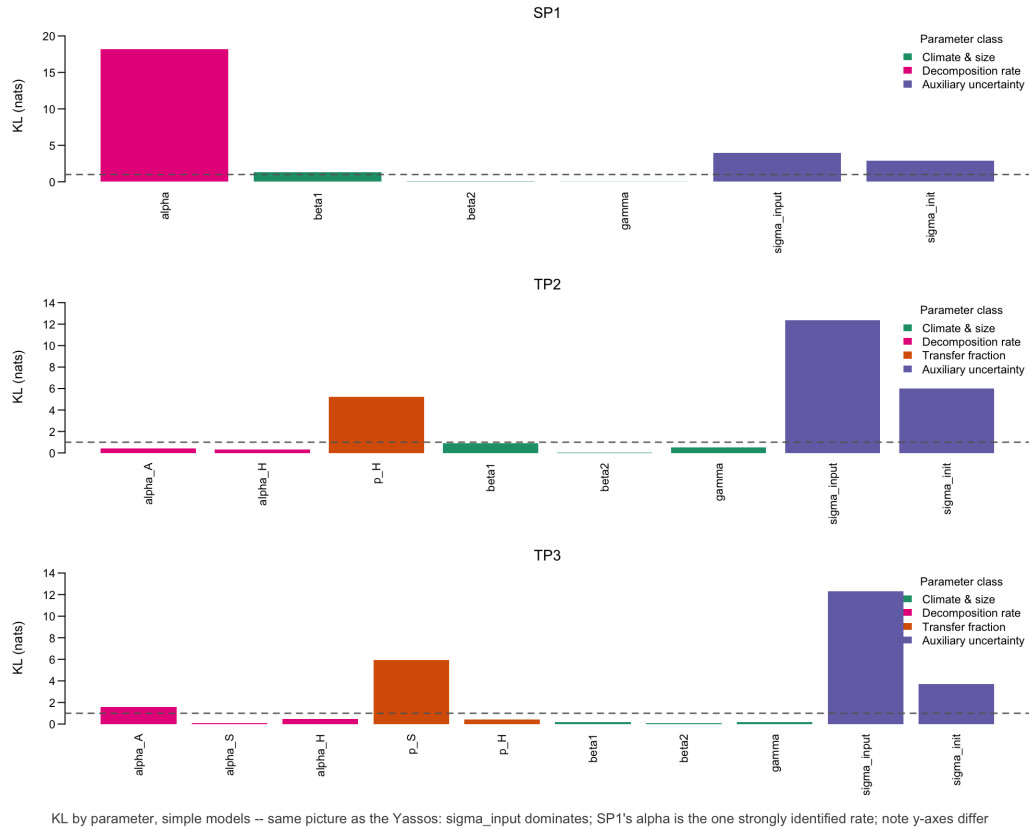


Figure 17: S4 — residual diagnostics. Residual-vs-fitted, normal Q-Q, and histogram of the log residuals, pooled over all six models (log is the multiplicative-normal error model’s natural scale). Roughly symmetric and near-Gaussian; the small negative offset (-0.115) is the systematic $\sim 11\%$ over-prediction — a likelihood property, not biogeochemistry.



KL by parameter, simple models — same picture as the Yassos: σ_{input} dominates; SP1’s alpha is the one strongly identified rate; note y-axes differ

Figure 18: S5 — information gain, the simple models. $KL(\text{posterior} \parallel \text{prior})$ for SP1/TP2/TP3 — the companion to Fig. 9. Same verdict: σ_{input} dominates; SP1’s single decomposition rate is the one strongly identified parameter in the study; TP2/TP3 rates gain ≈ 0 .

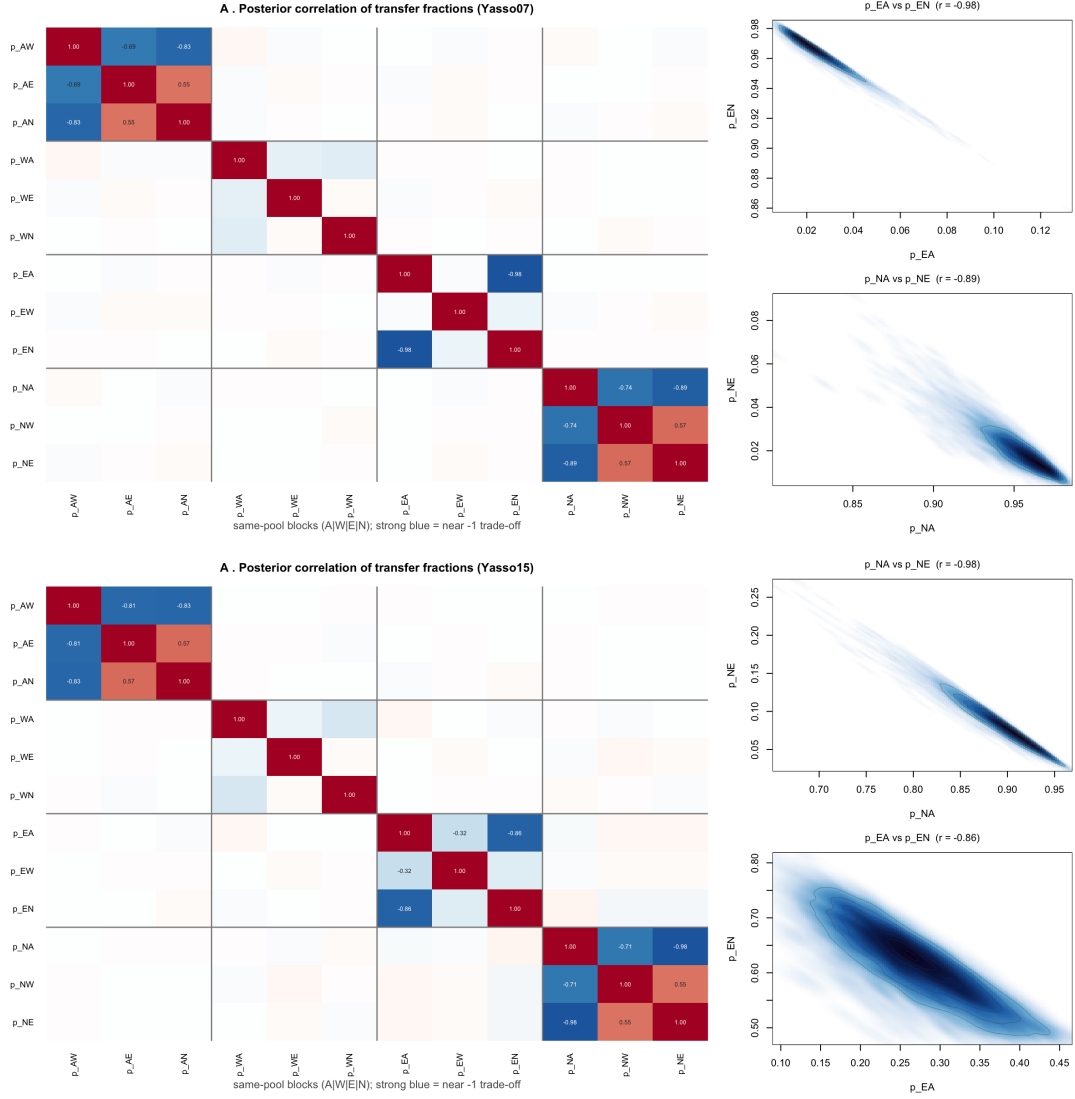


Figure 19: S6 — fraction-ridge structure, the other two Yassos. The same correlation heatmap and density ridges as Fig. 10, for Yasso07 (top) and Yasso15 (bottom). Identical block structure — so the non-identifiability is a property of the model class, not a single variant.

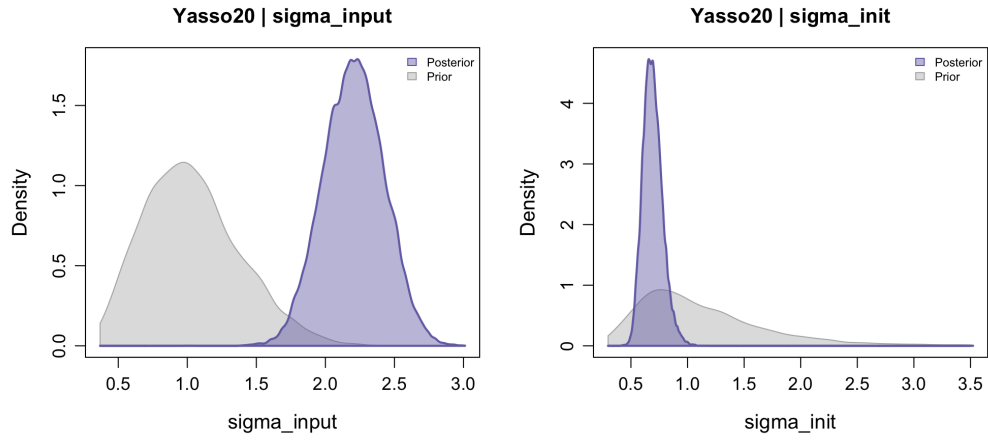


Figure 20: S7 — marginals for the identified auxiliary parameters. Posterior (purple) vs. prior (grey) for σ_{input} and σ_{init} (Yasso20). Both move decisively off the prior — the counterpoint to the non-identified transfer fractions: where the data *do* constrain a parameter, it shows. (More parameter marginals can be added.)